Course Code: 23BM101T

R23

PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)

I B.Tech I Semester Regular Examinations, January-2024

LINEAR ALGEBRA AND CALCULUS

(Common to all branches)

Time: 3 hours **Max. Marks: 70**

Note: i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.

ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.		Questions	BTL	CO	Marks
1.	a)	Find the rank of the matrix $A=\begin{pmatrix} 1 & 1 & 2 \\ 3 & 0 & 4 \\ 2 & 5 & -1 \end{pmatrix}$.	K3	CO1	2M
	b)	Solve the system of equations $3x-y=-1$ and $x+y=5$ if it is consistent.	K3	CO1	2M
	c)	Find an eigenvector corresponding to an eigenvalue $\lambda=5$ of a matrix A= $\begin{pmatrix} 5 & -10 & -5 \\ 2 & 14 & 2 \\ -4 & -8 & 6 \end{pmatrix}$	K3	CO2	2M
	d)	Let $A=\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$. Using Cayley-Hamilton theorem, find A^{-1}.	K3	CO2	2M
	e)	State the Rolle's theorem.	K3	CO3	2M
	f)	Is $f(x)=\lvert x-1\rvert$ satisfies Lagrange's Mean Value theorem on the interval $[0, 2]$?	K3	CO3	2M
	g)	If $u=\log_e\left(x^3+y^3+z^3-3xyz\right)$, find $\frac{\partial u}{\partial x}$.	K3	CO4	2M
	h)	If $x=r\cos\theta$ and $y=r\sin\theta$, find $\frac{\partial(x, y)}{\partial(r, \theta)}$.	K3	CO4	2M
	i)	Evaluate $\int_0^1\int_0^2\int_0^3 dx\,dy\,dz$.	K3	CO5	2M
	j)	Evaluate $\int_0^1\int_0^x y\,dy\,dx$.	K3	CO5	2M
		UNIT-I			
2.	a)	Find the rank of the matrix A= $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$ by reducing into normal form.	K3	CO1	5

	b)	Solve the following system of equations $$x+y+z=3$$ $$x+2y+3z=5$$ $$x+4y+7z=10$$ if the system is consistent.	K3	CO1	5M
		OR			
3.		Solve the following system of equations $2x+4y-z+w=7; x+y+4z-w=4; x+y+2z+2w=8; x+y+z-2w=1$ using Gauss elimination method.	K3	CO1	10M
		UNIT-II			
4.	a)	Find the eigenvalues and the corresponding eigenvectors of the matrix $A=\begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}$.	K3	CO2	5M
	b)	Let $A=\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$. Using Cayley-Hamilton theorem find A^5.	K3	CO2	5M
		OR			
5.		Reduce the quadratic form $x^2+3y^2+3z^2-2yz$ into a canonical form by using orthogonal transformation. Also discuss the nature of the given quadratic form.	K3	CO2	10 M
		UNIT-III			
6.	a)	Verify Lagrange's Mean Value theorem for the function $f(x)=\frac{x}{1+x}$ on $[1, 3]$.	K3	CO3	5M
	b)	Verify Rolle's theorem for the function $f(x)=\frac{2}{e^x+e^{-x}}$ on $[-1, 1]$.	K3	CO3	5M
		OR			
7.		Expand $f(x)=x^2\sin x$ as Taylor series about $x=\frac{\pi}{2}$ up to four terms.	K3	CO3	10M
		UNIT-IV			
8.		Verify the functions $u=\frac{x}{y-z}$, $v=\frac{y}{z-x}$ and $w=\frac{z}{x-y}$ are functionally dependent are not. If u, v and w are functionally dependent, then obtain the relation between them.	K3	CO4	10M
		OR			
9.		If A, B, C are the angles of a triangle, find the maximum value of $\cos A\cos B\cos C$.	K3	CO4	10M
		UNIT-V			
10.	a)	If A is the area of the rectangular region bounded by the lines $x=0$, $x=1$, $y=0$ and $y=2$, evaluate $\iint_A (x^2+y^2)\,dA$.	K3	CO5	5M
	b)	Evaluate $\int_0^1\int_0^{\sqrt{1-x^2}} y^2\,dy\,dx$ by changing the order of integration.	K3	CO5	5M
		OR			
11.		Evaluate $\iiint_R xyz\,dx\,dy\,dz$, where R is the region enclosed by the parabolas $z=x^2+y^2$ and $z=8-(x^2+y^2)$ using Cylindrical polar coordinates.	K3	CO5	10